

TRANSACTION MONITORING & FRAUD DETECTION FRAMEWORK

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1. PURPOSE AND SCOPE

This framework establishes the mandatory standards for transaction monitoring and fraud detection at Nordvik Bank ASA. It defines the system architecture, rule library, alert management workflow, investigation standards, escalation procedures, and quality assurance requirements that together constitute the Bank's first-level automated defence against money laundering, terrorist financing, fraud, and other financial crime conducted through the Bank's payments and account infrastructure.

This framework applies to all transaction channels including SWIFT, SEPA, domestic NOK transfers (BankAxept, Nets), card payments (Visa/Mastercard), ATM, mobile banking (Nordvik App), internet banking, and digital asset transactions. It applies to all customer segments: retail, SME, corporate, private banking, and correspondent banking.

This framework must be read in conjunction with the KYC/CDD Policy (NB-AML-KYC-001), the SAR Procedures (NB-AML-SAR-003), and the AML Risk Assessment Guidelines (NB-AML-RA-004).

2. REGULATORY BASIS

The transaction monitoring and fraud detection obligations of Nordvik Bank arise from the following primary regulatory sources:

Source	Obligation
Norwegian Money Laundering Act (2018) MLD	Monitoring of customer transactions and business relationships on an ongoing basis
EU 4AMLD/5AMLD, Article 14	Ongoing monitoring of business relationships; scrutiny of transactions inconsistent with knowledge
FATF Recommendation 10	Financial institutions required to conduct ongoing due diligence on the business relationship including
EBA Guidelines EBA/GL/2021/02	Risk-sensitive approach to monitoring; governance over automated systems
Norwegian FSA Circular 7/2018	Specific guidance on TMS calibration, false positive management, and investigator training
PSD2 / EBA Fraud Reporting (EBA/GL/2019/05)	Payment fraud monitoring and reporting obligations
Finanstilsynet AML Inspection Findings 2022	Requirement for documented rule rationale, tuning logs, and alert closure documentation

3. FRAMEWORK ARCHITECTURE

The Bank's transaction monitoring and fraud detection architecture operates across three integrated layers:

3.1 Layer 1 – Real-Time Fraud Prevention

Real-time controls operate at the point of transaction initiation, typically within 50–200 milliseconds. These controls are primarily designed to prevent fraud rather than detect it retrospectively. Real-time controls include:

- Card transaction authorisation scoring (ML-based, operating 24/7)
- Account takeover detection – device fingerprinting, behavioural biometrics, impossible travel
- Sanction screening at point of payment (SWIFT gpi, SEPA, Nets)
- Velocity checks – unusual frequency or volume of outgoing payments
- Duplicate payment detection
- Known fraud pattern matching (static blacklists and dynamic watchlists)
- Step-up authentication triggers for high-risk transactions

3.2 Layer 2 – Near-Real-Time Transaction Monitoring (NICE Actimize SAM)

The core transaction monitoring platform (NICE Actimize Suspicious Activity Monitoring) processes transactions in batch windows of 15 minutes to 4 hours. This layer applies rule-based detection across customer and counterparty transaction history and generates alerts for analyst review. The system ingests:

- All debit and credit transactions across all accounts and channels
- Customer reference data and KYC risk profile from the Core Banking System (Temenos T24)
- PEP and sanctions screening results
- Negative news and adverse media feeds
- Internal blacklists (known fraudsters, mule accounts)
- Historical alert dispositions and SAR filings

3.3 Layer 3 – Retrospective & Analytical Detection

Retrospective detection is conducted through scheduled analytical routines and investigator-driven deep-dive analysis. This layer identifies patterns that emerge over longer time horizons, including structuring (smurfing), layering across multiple accounts, and complex typologies that do not trigger Layer 1 or Layer 2 controls. Tools include:

- SQL-based analytical queries run weekly/monthly by Senior Analysts
- Network analysis tooling (i2 Analyst's Notebook) for relationship mapping
- Machine learning anomaly detection – unsupervised clustering of transaction behaviour
- Peer group analysis – comparing customer transaction patterns to similar customer cohorts
- Regulatory data requests and law enforcement production orders

3.4 System Integration Architecture

The monitoring systems are integrated with the Bank's core infrastructure as follows:

System	Role	Integration Method
Temenos T24 (Core Banking)	Source of transaction and customer data	Real-time API + batch file
NICE Actimize SAM	Primary AML transaction monitoring	API feed from T24; alert case management

Actimize RCM	Regulatory case management and SAR filing	Integrated with SAM
WorldCompliance / Refinitiv	PEP and sanctions data feeds	Daily update to screening engine
Proprietary ML Engine	Fraud scoring, anomaly detection	Containerised microservice; scores fed to SAM
i2 Analyst's Notebook	Network and relationship analysis	Export from SAM; manual import
Temenos Financial Crime Mitigation	Real-time card fraud prevention	Direct integration with Visa/MC
ServiceNow	Case ticketing and workflow management	API to SAM

4. RISK-BASED ALERT CALIBRATION

The Bank applies a risk-based approach to alert threshold calibration to ensure that monitoring intensity is proportionate to the ML/TF and fraud risk presented by each customer segment and transaction type. Thresholds are not set to a single universal standard but are differentiated by customer risk rating, product type, and jurisdiction.

4.1 Calibration Principles

- Alert thresholds must generate a volume of alerts that is manageable by the investigation team whilst ensuring no high-risk activity goes undetected
- False positive rate must be tracked, documented, and maintained within the Bank's accepted range (target: <85% false positive rate, measured monthly)
- All threshold changes must follow the change control procedure in Section 9.3
- Thresholds must be reviewed at minimum every 6 months and following any material change in the business, regulatory environment, or observed typologies
- Peer group analysis is used to calibrate 'normal' transaction behaviour for each customer segment, enabling anomaly detection relative to the peer group

4.2 Customer Risk Segment Thresholds

Monitoring thresholds are differentiated by customer risk rating. The following table shows indicative threshold multipliers relative to the standard (medium risk) baseline:

Risk Segment	Threshold Multiplier	Review Frequency	Example: Cash Alert Trigger
Low Risk	1.5x baseline	Annually	NOK 150,000 single cash transaction
Medium Risk (baseline)	1.0x	Semi-annually	NOK 100,000 single cash transaction
High Risk	0.5x	Quarterly	NOK 50,000 single cash transaction
Critical / PEP	0.25x	Monthly	NOK 25,000 single cash transaction
Correspondent Banking	Custom per entity	Quarterly	Per agreed monitoring profile

5. TRANSACTION MONITORING RULES – TYPOLOGY LIBRARY

The following rule library defines the core AML transaction monitoring rules deployed in NICE Actimize SAM. Each rule is referenced by a unique Rule ID. Full threshold parameters are documented in Annex A.

Rule ID	Rule Name	Typology	Description	Risk Level	Priority
AML-001	Large Cash Deposit	Cash Placement	Single or aggregated cash deposits exceeding threshold within 24-hour window. Alerts on	HIGH	P1
AML-002	Structured Cash Deposits (Smurfing)	Structuring	Multiple cash deposits below reporting threshold >=3 transactions within 5 business	HIGH	P1
AML-003	Rapid Funds Movement	Layering	Credit received and >=80% debited within 24 hours. Minimum transaction count: 3.	HIGH	P1
AML-004	Round Dollar Transactions	Structuring	Unusually high frequency of round-number transactions (divisible by 1,000 or 10,000). The	MEDIUM	P2
AML-005	High-Risk Jurisdiction Transfers	Cross-border ML	Outgoing wire transfers to FATF grey/black-listed countries or high-risk jurisdictions as per	HIGH	P1
AML-006	PEP Unusual Activity	Corruption / Bribery	Any transaction for PEP customers that deviates from their established profile by >200% i	CRITICAL	P1
AML-007	Shell Company Indicators	Layering / Integration	Incoming funds from entities with opaque structures, no web presence, or mismatched inc	HIGH	P2
AML-008	Trade Finance Anomaly	Trade-Based ML	Discrepancy between trade document value and payment amount >20%; mismatched cor	HIGH	P1
AML-009	Crypto Asset On/Off-Ramp	VASP ML	Transactions to/from identified cryptocurrency exchange wallet	MEDIUM	P2
AML-010	Dormant Account Activation	Account Takeover / ML	Account with no activity for >180 days receives large credit followed by rapid debit movem	HIGH	P1
AML-011	Salary Account Anomaly	ML Indicator	Retail salary account receives payments from multiple different business entities inconsis	MEDIUM	P2
AML-012	Charity / NPO Unusual Inflows	NPO Misuse	Non-profit organisation account receives unusually large or frequent payments inconsisten	HIGH	P2
AML-013	Real Estate Payment Pattern	Real Estate ML	Large transfers to solicitors, notaries, or property companies inconsistent with customer p	MEDIUM	P2
AML-014	Loan Repayment from Third Party	Third-Party Funds	Loan or credit facility repaid consistently from third-party accounts unrelated to the borrow	MEDIUM	P2
AML-015	Counter-flow Transactions	Back-to-Back	Customer sends and receives similar amounts to/from same counterparty within short time	HIGH	P1
AML-016	Tax Haven Transactions	Tax Evasion Indicator	Payments to/from known tax haven jurisdiction in combination with opaque entity structur	HIGH	P2
AML-017	Multiple Currency Exchanges	Currency Conversion ML	Multiple FX conversions within a short period of time across high-risk currency pairs.	MEDIUM	P2
AML-018	Terrorist Financing Indicators TF		Payments to jurisdictions, entities, or individuals associated with known TF typologies. Lin	CRITICAL	P1

6. FRAUD DETECTION RULES & CONTROLS

Fraud detection operates across both real-time prevention controls and near-real-time monitoring. The Bank's fraud taxonomy distinguishes between external fraud (perpetrated against the Bank or its customers by third parties) and internal fraud (perpetrated by employees or insiders).

6.1 External Fraud Rule Set

Rule ID	Fraud Type	Description	Detection Method	Action
FRD-001	Card Not Present (CNP) Fraud	Unauthorized online/telephone card transactions	Transaction USR, device risk, IP, shipping, step up, last, matching velocity	Block card, decline, step up, last, matching velocity
FRD-002	Account Takeover (ATO)	Fraudster gains access to customer account	Account takeover, IP, device, social engineering, data transfer, flag for possible	Block account, step up, last, matching velocity
FRD-003	Authorized Push Payment (APP) Fraud	Customer manipulated into authorising payment to fraudster	Payment to high risk list, Real time payment go 2 step, first payment	Block payment, step up, last, matching velocity
FRD-004	Cheque Fraud	Counterfeit, altered, or stolen cheques	Check fraud, ML model, value, delay, clearing, deal, all over	Block payment, step up, last, matching velocity
FRD-005	Identity Fraud	Use of stolen or fabricated identity to open account	IP, Canada, social, on credit, Application, cross check, level, test, fraud, consider	Block account, step up, last, matching velocity
FRD-006	Money Mule Activity	Account being used by third parties to receive funds	Rapid funds in, out, payment, reference, structure, fraud, investigation, report	Block account, step up, last, matching velocity
FRD-007	First Party Fraud	Customer deliberately misrepresents information	Credit application, inconsistent, credit, frequency, investigation, early default	Block account, step up, last, matching velocity
FRD-008	Phishing / Vishing	Fraudster impersonates Bank to extract customer information	Customer reports, large payment, follow up, usual, notification, incident	Block account, step up, last, matching velocity
FRD-009	Investment Scam	Customer deceived into investing in fraudulent scheme	Large outgoing transfers to unknown, warning message, platform, social, investment	Block account, step up, last, matching velocity
FRD-010	Business Email Compromise (BEC)	Fraudster intercepts or spoofs corporate email	Change to direct, legitimacy, payment, verification, call the customer, large payment, platform	Block account, step up, last, matching velocity

6.2 Internal Fraud Controls

The Bank applies specific controls to detect potential internal fraud and employee misconduct:

- Privileged access monitoring – alerts on unusual system access patterns, especially outside business hours
- Transaction override monitoring – alerts when employees override system controls
- Customer account access logging – alerts on staff accessing accounts of family members, themselves, or VIP customers without documented business purpose
- Fee waiver and adjustment monitoring – unusual frequency or value of fee waivers applied by individual staff members
- Whistle-blower reports – dedicated confidential reporting channel; all reports investigated by Internal Audit
- Segregation of duties controls – enforced in core banking; no single user can initiate and approve a payment above NOK 50,000

7. ALERT MANAGEMENT & INVESTIGATION WORKFLOW

All alerts generated by the transaction monitoring system are managed through a structured workflow designed to ensure consistent, documented, and timely investigation.

7.1 Alert Triage & Assignment

Upon alert generation, the system automatically assigns a Priority level (P1–P4) based on the triggering rule, customer risk rating, and transaction amount. Priority levels determine the investigation SLA:

Priority	Description	SLA (Business Hours)	Example Triggers
P1 – Critical	Immediate risk of financial loss or regulatory breach	4 hours	Sanctions match; suspected TF; active fraud; ATO confirmed
P2 – High	Significant ML/fraud risk requiring prompt attention	24 hours	PEP unusual activity; high-risk jurisdiction transfer; APP fraud
P3 – Medium	Moderate risk; pattern of concern	72 hours	Structuring pattern; dormant account activation; salary anomaly
P4 – Low	Low-risk indicator; bulk processing	5 business days	Round-dollar transactions; low-value velocity alerts

7.2 Investigation Standards

Every alert investigation must result in a documented case file that includes the following elements:

- Summary of the triggering alert(s) and rule(s) fired
- Customer profile overview including KYC risk rating, onboarding date, and last review date
- Review of all transactions for the alert period (minimum 90 days lookback for P1/P2; 30 days for P3/P4)
- Analysis of counterparty information including known relationships and risk indicators
- Narrative explanation of the observed activity and its assessment against known typologies
- Assessment of whether the activity is consistent with the customer's known profile and declared purpose
- Information-gathering steps taken (customer contact, third-party data, public domain checks)
- Disposition decision with documented rationale
- If SAR filed: SAR reference number and key grounds for suspicion
- Investigator name, date, and time of completion; supervisor review sign-off for P1/P2 cases

7.3 Customer Contact During Investigation

Analysts may contact customers to request clarification of unusual transactions. The following rules apply:

- Customer contact must not be initiated where it would tip off the customer to a potential SAR filing (tipping-off offence under the Money Laundering Act)
- Where a SAR has been filed or is under consideration, all customer contact must be approved by the MLRO
- Contact must be documented in the case file with date, time, channel, and content of communication
- Contact must be conducted by the designated investigator or relationship manager under Compliance direction – not by the customer's primary RM without Compliance approval

8. ESCALATION & DISPOSITION

8.1 Alert Disposition Codes

Each investigation must be closed with a documented disposition code (see Annex B for full list). Primary disposition codes:

Code	Description	Action Required
D01 – No Suspicious Activity	Alert reviewed; activity fully explained by customer	Close alert, document business purpose
D02 – Unusual but Explainable	Activity is unusual but a satisfactory explanation has been obtained	CDD refresh if profile change identified
D03 – CDD Refresh Required	Activity has revealed gaps in KYC; no SAR required	Edge CDD refresh, update alert pending update
D04 – Escalated to MLRO	Activity cannot be satisfactorily explained; suspicious internal STR; await MLRO decision on SAR	File internal STR; await MLRO decision on SAR
D05 – SAR Filed	MLRO has determined that a SAR should be filed	File SAR with Finstilsynet; await consent if required
D06 – Account Restriction	Immediate risk of harm requires account restriction	Responding investigation MLRO; escalate to Head of FC
D07 – Law Enforcement Production	Alert relates to an active law enforcement enquiry	Preserve records; comply with order; do not tip off customer
D08 – False Positive – System Alerting	System consistently generating false positives; rule engine review	Log false positives; escalate to Model Governance team

8.2 Escalation to MLRO

An internal Suspicious Transaction Report (STR) must be escalated to the Money Laundering Reporting Officer (MLRO) where:

- The investigator has formed a suspicion, or has reasonable grounds to suspect, that a transaction involves the proceeds of crime or is connected to terrorist financing
- The activity cannot be satisfactorily explained by the customer's known profile and circumstances
- The customer has refused to provide clarification or documentation requested
- A pattern of alerts over time suggests a systemic issue
- A law enforcement agency has made a production order or informal enquiry relating to the customer

The MLRO must review all internal STRs within 5 business days (P1: same business day) and determine whether to file a Suspicious Activity Report (SAR) with the Financial Intelligence Unit (Finanstilsynet / ØKORIM).

9. MODEL GOVERNANCE

The transaction monitoring and fraud detection models are subject to the Bank's Model Risk Management Policy (NB-RISK-MRM-007). Specific requirements for financial crime models include:

9.1 Model Validation

- All new detection models and rules must be validated by the Model Risk function prior to deployment
- Validation includes: retrospective testing on 12 months of historical data; sensitivity analysis; false positive/negative rate analysis; peer comparison
- Validation results must be documented and approved by the Model Risk Committee
- Independent validation by a third party is required for models classified as 'High' or 'Critical' under the MRM Policy

9.2 Ongoing Performance Monitoring

Metric	Target	Frequency	Owner
False Positive Rate	<85%	Monthly	Head of Financial Crime
Alert Volume (vs. baseline)	±20% of baseline	Weekly	TM Operations Manager
Alert Closure Rate (within SLA)	≥95%	Monthly	TM Operations Manager
SAR Filing Rate (% of alerts)	Tracked; no specific target	Monthly	MLRO
Detection Rate (known fraud)	≥90% of confirmed fraud cases	Quarterly	Model Risk
Rule Coverage (vs. FATF typologies)	100% coverage of major typologies	Semi-annually	Head of FC

9.3 Rule Change Control

All changes to detection rules, thresholds, and models must follow the change control procedure:

- Change request submitted by TM Operations or Financial Crime to Model Risk team
- Impact assessment: estimated change in alert volume, false positive rate, coverage
- Approval: routine changes – Head of FC; threshold changes >20% – MLRO; new models – Model Risk Committee
- Implementation in staging environment with parallel running for minimum 2 weeks before production deployment
- Post-implementation review after 30 days
- All changes logged in the Rule Change Register maintained by Model Risk

10. REPORTING & MI

The following management information (MI) reports are produced and distributed on the frequencies noted:

Report	Frequency	Recipients
Daily Alert Volume & SLA Adherence Dashboard	Daily	Head of FC; TM Operations Manager
Weekly Financial Crime Activity Report	Weekly	CCO; Head of FC; Senior Management
Monthly TM Performance Report (FP rates, model metrics)	Monthly	CCO; Model Risk Committee; BRCC
Quarterly Financial Crime Risk Report	Quarterly	Board BRCC; Executive Committee
Annual Fraud Losses & Recovery Report	Annual	Board; Executive Committee; Auditors
Regulatory SAR/STR Statistics	Monthly (to FSA on demand)	MLRO; CCO
Model Performance Dashboard	Monthly	Model Risk; Head of FC

11. QUALITY ASSURANCE

The Quality Assurance (QA) function within the Financial Crime Prevention team conducts ongoing review of alert investigations and dispositions to ensure consistency, completeness, and accuracy.

- **Case File QA Sampling:** 10% of all closed P3/P4 cases; 25% of P2 cases; 100% of P1 cases and all SAR-related cases are subject to QA review.
- **QA Criteria:** Timeliness (SLA adherence); completeness (all required elements present); accuracy (analysis is correct and supported by evidence); consistency (comparable cases are treated consistently); documentation quality.
- **QA Findings:** QA findings are logged and reported to the Head of FC monthly. Recurring errors trigger mandatory retraining. Systemic failures are escalated to the CCO.
- **External QA:** Independent QA of the TM function is conducted by Internal Audit annually and may be supplemented by external expert reviews commissioned by the CCO.

ANNEX A FULL RULE INVENTORY & THRESHOLDS (EXTRACT)

RESTRICTED: Specific threshold values are classified as Internal Restricted. Access to the full threshold table in the Rule Change Register requires Head of FC approval.

Rule ID	Rule Name	Base Threshold (NOK)	High-Risk Threshold (NOK)	Lookback Window	Min. Transactions
AML-001	Large Cash Deposit	100,000	50,000	24 hours	1
AML-002	Structured Cash (Smurfing)	80,000 aggregate	40,000 aggregate	5 business days	3
AML-003	Rapid Funds Movement	200,000	100,000	24 hours	3 debits
AML-005	High-Risk Jurisdiction Transactions	25,000	10,000	Per transaction	1
AML-010	Dormant Account Activation	50,000	25,000	Per transaction	1
AML-015	Counter-flow Transactions	75,000 each leg	40,000 each leg	5 days	2
FRD-001	CNP Fraud	ML score	0.8 threshold	Real-time	1
FRD-003	APP Fraud	First payment to new App account	Any amount	Real-time	1
FRD-006	Money Mule	20,000 outflow	10,000 outflow	48 hours	3 inflows